

Piecewise Linear Regression

Optimization problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^n$.

Equivalent linear programming:

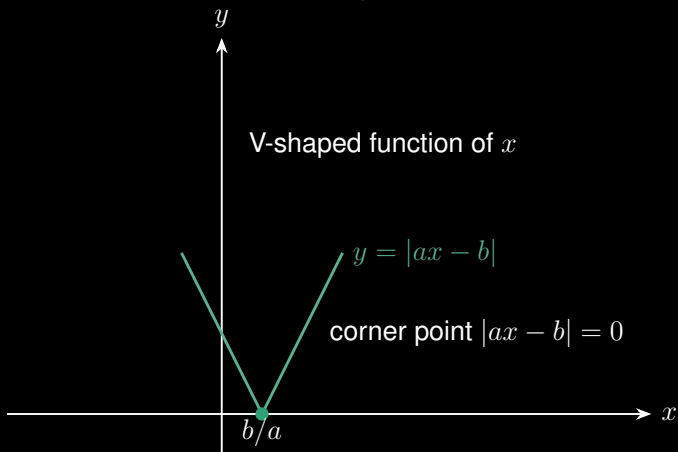
$$\begin{aligned} \min_{\mathbf{x}, t_1, \dots, t_n} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & -t_i \leq a_i \mathbf{x} - b_i \leq t_i, \quad \forall i \end{aligned}$$

Piecewise Linear Regression

How to minimize $\sum_{i=1}^n |a_i x - b_i|$ without using linear programming?

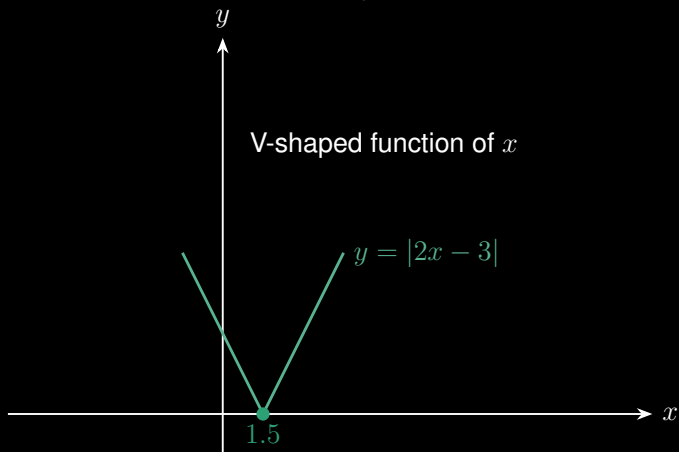
Piecewise Linear Regression

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$



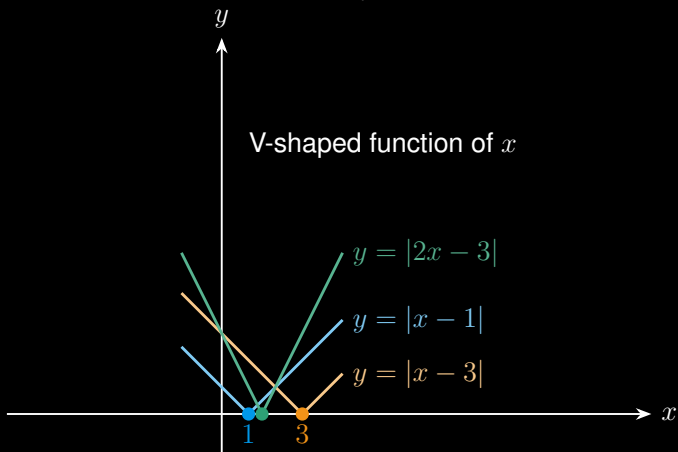
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$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$



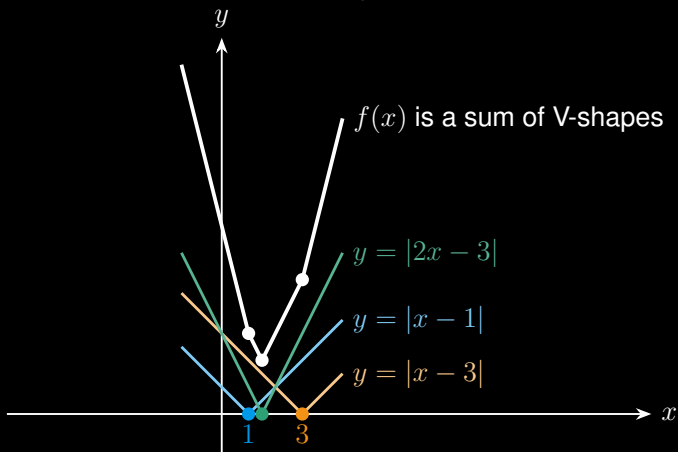
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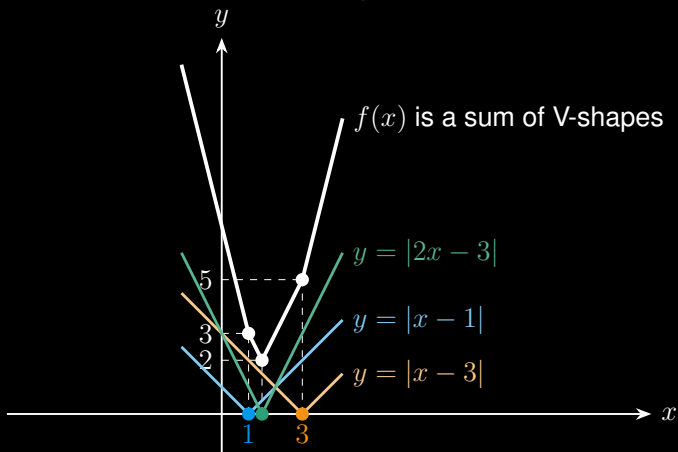
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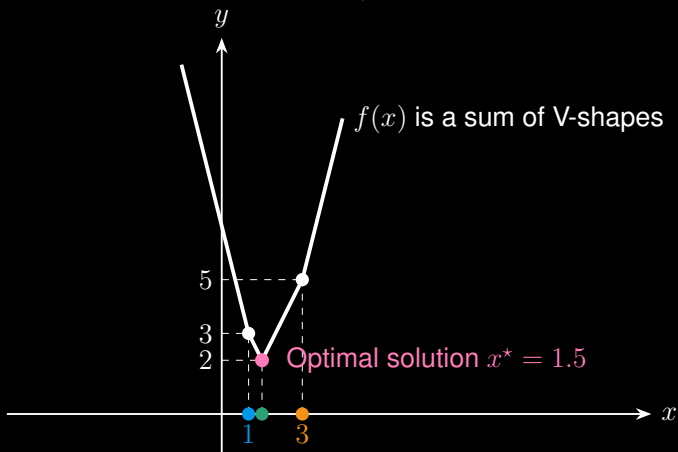
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Piecewise Linear Regression without Using Linear Programming

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Data pairs:

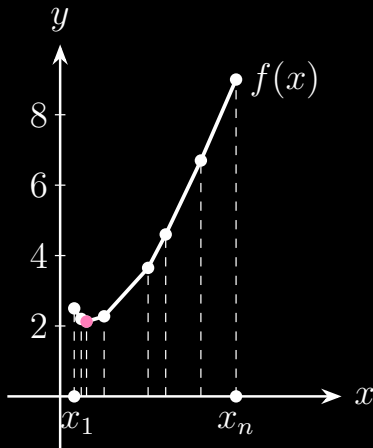
$$a_i \in \{0.4, 0.5, 0.1, 0.2, 0.4, 0.1, 0.1, 0.2, 0.5\}$$

$$b_i \in \{0.5, 0.2, 0.3, 0.5, 0.3, 0.5, 0.4, 0.5, 0.3\}$$

Algorithm:

- ① Compute all points $x_i = b_i/a_i$;
- ② Sort values $x_1 \leq x_2 \leq \dots \leq x_n$;

$$x_i \in \{0.4, 0.6, 0.75, 1.25, 2.5, 2.5, 3, 4, 5\}$$



Piecewise Linear Regression without Using Linear Programming

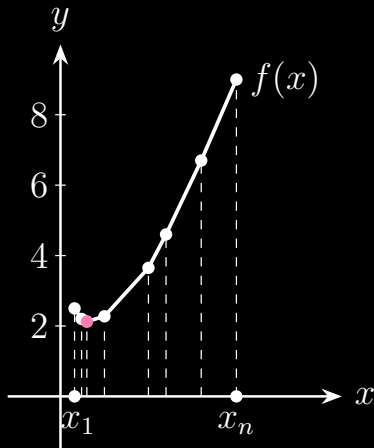
$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Algorithm:

- ① Compute all points $x_i = b_i/a_i$;
- ② Sort values $x_1 \leq x_2 \leq \dots \leq x_n$;
- ③ Let $k = \lfloor \frac{n}{2} \rfloor$ and compute the slope

$$\frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$

- ④ Find some x_ℓ such that the slope is negative on $[x_{\ell-1}, x_\ell]$ but positive on $[x_\ell, x_{\ell+1}]$.



Thanks for your attention!

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